Quantitative Investment Project

October 14, 2026

1 Introduction

The project is articulated in four main sections. Section 2 is for the design of a Basket to hedge a large stock exposure and section 3 compares two different approaches for portfolio allocation with risk control, while Section 4 is a mere variation variation of Section 3. Section 1 will be implemented in a straightforward way by adapting the sample python code that we used in class for the Baby Portfolio Microsoft-Starbucks case (CVXOPT optimizer) and Section 3 will be implemented using the starter code provided in the appendix of this document (CVXPY optimizer). Section 4 is an extension of Section 3 to the case when we add limits in the exposure (weights) and section is an attempt to connect these strategies to more realistic situations. Note that in this baby project, you will use the sample estimators: sample mean and sample covariance to build the optimizers' inputs[1].

2 Designing a Basket to Beta Hedge a Large Stock Exposure

An investor wishes to design a hedge basket to offset of a position she holds in stocks of Apple (Ticker: AAPL) by taking position in a Mean-variance portfolios with well-chosen target betas. The market (S&P-500 index) will be represented by the SPY ETF and the investment universe she is considering is a set 10 ETFs with tickers below:

$$\mathcal{U} = \{\text{FXE}, \text{EWJ}, \text{GLD}, \text{QQQ}, \text{SHV}, \text{DBA}, \text{USO}, \text{XBI}, \text{ILF}, \text{EPP}, \text{FEZ}\}.$$

To hedge the exposure, the investors uses the two fund theorem that says that solutions of mean-variance type optimization problems combine linearly, The two-fund theorem states that if $\omega_{mv}(\beta_T)$ is the solution of the problem

$$\text{P}_{mV}(\beta_T): \begin{cases} \min\limits_{\omega} & \omega^T \Sigma \omega \\ \text{s.t.} & \beta^T \omega = \beta_T \text{ and } \mathbf{e}^T \omega = 1, \text{ where } \mathbf{e} = (1, 1, \ldots, 1)^T \end{cases} \tag{1}$$

where β_T is the Target Beta, then given any two different targets β_a and β_b and their respective optimal portfolios $\omega_{mv}(\beta_a)$ and $\omega_{mv}(\beta_b)$, we have:

$$\beta_T = \alpha\beta_a + (1-\alpha)\beta_b \implies \omega_{mv}(\beta_T) = \alpha\omega_{mv}(\beta_a) + (1-\alpha)\omega_{mv}(\beta_b)$$

As a preliminary step, download the historical data for the ticker AAPL, the SPY and the tickers in the set $\mathcal{U}$ for the period starting in Jan. 1st, 2025 and ending on Feb. 28th, 2026. Remove all missing entries. Compute and store the daily returns of all securities in a dataFrame.

[1] Your are required to use convex optimizers in this project: solutions using nonlinear optimizers will not be accepted.

1. **Two-fund Theorem with Target Beta.** In this question, you will assume that the investment decision is taken on January 2nd, 2026 and that the set $\mathcal{U}$ is considered as your investment.

   - Use 1 year of historical data to compute the sample Expected Returns ρ, sample Covariance Σ and individual betas β_i's for each security in the investment universe $\mathcal{U}$. The input vector β of the portfolio Betas has components $\beta_i = \dfrac{cov(r_i, r_M)}{\sigma^2(r_{SPY}}$ where each β_i is the Beta of security S_i with respect to the market (S&P 500) represented by the SPY.
   - Given $\beta_a = 0.5$, $\beta_b = 1.5$ and $\beta_T = 1$, find α such that $\beta_T = \alpha\beta_a + (1-\alpha)\beta_b$ and apply the 2 fund-theorem to derive the portfolio $\omega_{mv}(\beta_T)$ from $\omega_{mV}(\beta_a)$ and $\omega_{mV}(\beta_b)$.
   - Use CVOPXT optimizer to solve directly $\mathrm{P}_{mV}(1)$) and compare the result with the one from the previous item.

2. **Hedging a Long position in Apple.** In this question, you will still assume that the investment decision is taken on January 2nd, 2026 and you will use 1 year of past historical data to compute the Expected Returns, Covariance and Betas needed for the optimization.

   (a) Compute the betas of the securities with respect to the S&P-500 index. Compute β_A, the Beta of the stock Apple with respect to the S&P-500 index.

   (b) **Beta Hedging Portfolio.** Find the portfolio $\omega_{mV}(\beta_A)$. How would you use that portfolio to hedge your position in Apple? What is the expecting return of your hedged strategy?

   (c) **Beta Neutral Portfolio.** Considering the enlarged investment universe $\mathcal{V} = \mathcal{U} \cup \{AAPL\}$, find in that universe the Minimum Variance Portfolio with Target Beta equal to 0. (Beta neutral Portfolio). Compare that portfolio with the aggregated position you built in the question 2(b). How do they compare in terms of Risk, Expected Return and realized Beta?

3. **Comparison of the 2 Hedging strategies**

   (a) Compute the realized daily returns of the Portfolios built in questions 2(a) and 2(b) for the period starting in January 3rd, 2026 and ending on February 28th 2026.

   (b) Compare the time series of the realized returns of the 2 hedging strategies: probability distribution, graphs of cumulative returns, mean, volatility, Sharpe ratio, Sortino ratio, 95%-Value-at-Risk (VaR) , 95%-Conditional Value-at-Risk (CVaR), Skewness, Kurtosis and realized Beta.

4. **Chasing Performance and dynamic Hedging**

   (a) Can you improve the PnL of the Hedged strategy by adding a linear term $-\lambda\rho^T\omega$ to the Objective function, where ρ is the Expected Return and $\lambda > 0$?

   (b) Can you improve the hedge by using 60 days to compute the betas and covariance and re-hedge every week?

3 Comparison with an Risk Control Strategy

Understanding beta as a measure of market risk, one may think of a portfolio with target beta =1 as a portfolio replicating the market risk profile. In this section, we want to compare the behavior of two

different approaches: setting target beta at a maximum of 1 (T_β strategy) and controlling directly the volatility (T_{var} strategy). To that end, you will implement the two following strategies:

$$\mathrm{T}_\beta(\lambda): \begin{cases} \min\limits_{\omega} & \omega^T\Sigma\omega - \lambda\rho^T\omega, \text{ for } \lambda > 0, \\ \text{s.t.} & \beta^T\omega \leq 1 \text{ (Limit Beta)} \\ & \mathbf{e}^T\omega = 1, \end{cases} \quad \text{use CVXOPT for the coding} \tag{2}$$

and

$$\mathrm{T}_{var}(\lambda): \begin{cases} \min\limits_{\omega} & \omega^T\Sigma\omega - \lambda\rho^T\omega, \text{ for } \lambda > 0, \\ \text{s.t.} & \omega^T\Sigma\omega \leq \sigma^2_{SPY} \text{ (Market risk)} \\ & \mathbf{e}^T\omega = 1, \end{cases} \quad \text{use starter CVXPY code from appendix} \tag{3}$$

Use 60 rolling windows to days to estimate the expected returns ρ, covariance matrix matrix Σ, betas β and the SPY variance σ^2_{SPY}. Reallocate the weights of each strategy on a weekly basis the period starting in January 3rd, 2026 and ending on February 28th 2026. Consider the case where $\lambda = 2$.

1. Compute the realized daily returns of the Portfolios built using the T_β strategy and the volatility T_{var} strategy for the period starting in January 3rd, 2026 and ending on February 28th 2026.
2. Compare the 2 hedging strategies in the same way as in 3(b) from Section 2 above.
3. Provide your recommendations as for which strategy you would recommend and why.
4. Repeat the analysis in items 1 to 3 using 1 year of look-back for the covariance and the betas.
5. Repeat the analysis in items 1 to 3 using 5 years of look-back for the covariance and the betas.

What are your overall conclusions for this section.

4 Controlling Exposure and leverage

In section, analyze the impact of adding limits on leverage and exposure. To avoid large swing in weights, consider again the T_β strategy and the T_{var} strategy while adding constraints on the weights for a better control of the size of the holdings:

$$\mathrm{T}^\delta_\beta(\lambda): \begin{cases} \min\limits_{\omega} & \omega^T\Sigma\omega - \lambda\rho^T\omega, \text{ for } \lambda > 0, \\ \text{s.t.} & \beta^T\omega \leq 1 \text{ (Limit Beta)} \\ & \mathbf{e}^T\omega = 1, \ -\delta \leq \omega_i \leq \delta, \ \delta \in \mathbb{R} \end{cases} \quad \text{use CVXOPT for the coding} \tag{4}$$

and

$$\mathrm{T}^\delta_{var}(\lambda): \begin{cases} \min\limits_{\omega} & \omega^T\Sigma\omega - \lambda\rho^T\omega, \text{ for } \lambda > 0, \\ \text{s.t.} & \omega^T\Sigma\omega \leq \sigma^2_{SPY} \text{ (Market risk)} \\ & \mathbf{e}^T\omega = 1, \ -\delta \leq \omega_i \leq \delta, \ \delta \in \mathbb{R} \end{cases} \quad \text{use starter CVXPY code from appendix.} \tag{5}$$

Choosing the best look-back period from section 3 (justify your choice), compare the T^δ_β and the T^δ_{var} strategies for $\delta = 0.15$, $\delta = 1$ and $\delta = 2$.

5 Towards a more real(istic) approach

Actually, investors would more interested in maximizing the expected return under volatility constraint. To implement such a strategy while keeping some robust properties, one might consider the following regularized problem which can be solved using the same CVXPY code:

$$\mathrm{T}^{\varepsilon}_{var}: \begin{cases} \min\limits_{\omega} & \omega^T(\varepsilon_r * I_d^n)\omega - \rho^T\omega, \\ \text{s.t.} & \omega^T\Sigma\omega \leq \sigma^2_{SPY} \text{ (Market risk)}, \ \mathbf{e}^T\omega = 1, \end{cases} \tag{6}$$

where $I_d^n = \mathrm{diag}(\mathbf{e})$ is the identity matrix in $\mathbb{R}^n$ and $\epsilon_r = 10^{-4}$ is a ridge regularization coefficient.[2] Now consider the regularized Target Limit Beta problem

$$\mathrm{T}^{\varepsilon_b}_{\beta}: \begin{cases} \min\limits_{\omega} & \omega^T(\Sigma + \varepsilon_b * I_d^n)\omega - \rho^T\omega, \\ \text{s.t.} & \beta^T\omega \leq 1 \text{ (Limit Beta)}, \ \mathbf{e}^T\omega = 1, \end{cases} \tag{7}$$

where ϵ_b is a ridge regularization coefficient taking values in the set

$$\mathcal{S}_r = \left\{10^{-4}, 0.5\times10^{-3}, 10^{-3}, 10^{-2}, 0.5\times10^{-2}, 10^{-1}, 0.5\times10^{-1}, 1\right\}.$$

Using 60 days of look-back to estimate the covariance matrix, betas and expected returns, compare the portfolios resulting from weekly rebalancing of T^{ε}_{var} and $T^{\varepsilon_b}_{\beta}$ for ε_b taking increasing values in $\mathcal{S}_r$. Pay special attention to the distribution of returns, the tail risk (VaR and CVaR), the magnitude of the weights and the cumulative PnL. What trend do you observe and what would you recommend to the investors interested in implementing these strategies?

6 Validation

Use your best strategy from the previous section to run a report the results of a backtest with weekly rebalancing covering the period the period from January 2021 to end of September 2026. Share your analysis of the results.

7 Final Thoughts

What are your overall conclusions, lessons learned and recommendations from this project?

Nota Bene:

- **I am expecting your own conclusions and not some copy and paste from Claude, GPT or Gemini.**
- **If you are using GenAI, please disclose it clearly and be ready to defend and explain your work during the presentation!**

[2] note that $\omega^T(\varepsilon_r * I_d^n)\omega = \|w\|_2^2$.

8 Appendix: Sample code for section T_{var} strategy

The sample code is almost ready to use, but make sure to add properly the specified constraints.

```
import cvxpy as cp
import numpy as np

# Problem formulation
# Minimize x.T * P * x + q.T * x
# Subject to x.T * Sigma * x <= sigma_val, sum (components of x) =  1

# Problem data (example values)
n = 2 # Dimension of x (dimension of the investment universe)
P = np.array([[2., 0.], [0., 2.]]) # Semi-Definite Positive matrix  (Covariance)
q = np.array([1., 2.])             # Linear term  (q = - lambda * Exp Return)
Sigma = np.array([[1., 0.], [0., 1.]]) #  (Covariance matrix)
sigma_val = 1                  # Variance upper limit (Market variance)

# Define the optimization variable (optimal allocated portfolio weights)
x = cp.Variable(n)

# Define the objective function
objective = cp.Minimize(0.5 * cp.quad_form(x, P) + q.T @ x)

# Define the constraints (must be a list)
# Use cp.quad_form for the quadratic constraint
# Ad sum of components of x equals 1
constraints = [
    cp.quad_form(x, Sigma) <= sigma_val,
    cp.sum(x) == 1
]

# Create the problem object
problem = cp.Problem(objective, constraints)

# Solve the problem
problem.solve()

# Print the results
print(f"Status: {problem.status}")
print(f"Optimal value: {problem.value}")
print(f"Optimal x: {x.value}")
```